

Week 04

Complex SQL Queries

Cảm ơn thầy Trần Duy Quang đã cung cấp template cho môn học
Cảm ơn thầy Hồ Tuấn Thanh và thầy Đỗ Nguyên Kha đã xây dựng nội dung thực hành



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Notes

Put **ALL** SQL statements to **ONE** text (SQL) document names student ID 1651xxx_W04_yy.sql (yy is the number of solved questions):

- Comment Student ID and Full name on top of the file
- Run ALL lines in the file without any error
- USE the CORRECT database

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SQL conditions

SQL conditions can be simple comparison: **>**, **<**, **=** (not **==**), **>=**, **<=**, **<>** (not **!=**) or check **IS NULL** or **IS NOT NULL** or set operation **IN**, **NOT IN**, **EXISTS**, **NOT EXISTS** or pattern matching **LIKE**. SQL conditions can be combined by logical operators **AND**, **OR**, and **NOT**.

Example:

Retrieve all employees (ID, FirstName, LastName) who are female and have no manager to report

```
SELECT ID, FirstName, LastName FROM Employees WHERE Sex = 'F' AND ReportID IS NULL  
AND DepartmentID IN (4, 5) AND FirstName LIKE 'S%'
```

DISTINCT

SELECT DISTINCT eliminates duplicates rows

Example:

```
SELECT ReportID FROM Employees WHERE ReportID IS NULL
```

```
SELECT DISTINCT ReportID FROM Employees WHERE ReportID IS NULL
```

UNION

UNION combines 2 data sets and eliminates duplicates rows

Example:

```
SELECT ID FROM Employees WHERE ReportID = 2  
UNION  
SELECT ID FROM Employees WHERE DepartmentID = 4
```

UNION ALL combines 2 data sets and doesn't eliminate duplicates rows

EXCEPT

EXCEPT return rows from data set 1 what doesn't exists in data set 2 and eliminates duplicates rows

Example:

```
SELECT ID FROM Employees WHERE ReportID = 2  
EXCEPT  
SELECT ID FROM Employees WHERE DepartmentID = 4
```

INTERSECT

INTERSECT returns rows that appears in both data sets and eliminate duplicates rows

Example:

```
SELECT ID FROM Employees WHERE ReportID = 2
INTERSECT
SELECT ID FROM Employees WHERE DepartmentID = 4
```

Nested query

SQL query can be used as a parameter of a condition of another query

Example:

Retrieve all employees (ID, FirstName, LastName) who has the same manager as John Smith

```
SELECT ID, FirstName, LastName FROM Employees
WHERE ReportID IN (SELECT ReportID FROM Employees WHERE FirstName = 'John' AND
LastName = 'Smith')
```

Other comparisons can be used: = **ANY** (or = **SOME**). Other operators >, <, >=, <=, <> can be combined with **ANY** or **SOME** or **ALL**

Correlated Nested Queries

When an attribute of the **outer query** appears in the nested query, the nested query is called correlated.

Example:

Find all employees (ID, FirstName, LastName) who are managers of a department

```
SELECT e.ID, e.FirstName, e.LastName
FROM Employees AS e
WHERE EXISTS (SELECT * FROM Departments AS d WHERE d.ManagerID = e.ID)
```

1. Lab

A: YY: 07 (7 queries)

Database name: Company (Ref: Fundamentals of database systems)

Tables:

- Employees:
 - ID: int, not null, primary
 - Code: char(9), not null, unique
 - FirstName: nvarchar(50), not null
 - LastName: nvarchar(50), not null
 - Birthday: date
 - Sex: char, not null ('F', 'M', 'N'). N is non-disclosure
 - Address: varchar(255)
 - Salary: int
 - SupervisorID: int, report to employee id, self-reference
 - DepartmentID: int, reference to the ID of Departments table
- Departments:
 - ID: int, not null, primary
 - Name: nvarchar(50), not null
 - ManagerID: int, reference to the ID of Employees table
 - ManagerStartDate: date
- DepartmentLocations:
 - DepartmentID: int, not null, primary, reference to the ID of Departments table
 - Location: nvarchar(50), not null, primary
- Projects:
 - ID: int, not null, primary
 - Name: nvarchar(50), not null
 - Location: nvarchar(50), not null
 - DepartmentID: int, reference to the ID of Departments table
- EmployeeProjects:
 - EmployeeID: int, not null, primary, reference to the ID of Employees table
 - ProjectID: int, not null, primary, reference to ID of Projects table
 - Hours: float
- Dependents:
 - EmployeeID: int, not null, primary, reference to the ID of Employees table
 - Name: nvarchar(50), not null, primary
 - Sex: char, not null ('F', 'M', 'N'). N is non-disclosure
 - Birthday: date
 - Relationship: nvarchar(50), not null

Create SQL file with requirements:

1. Create database
2. Create tables
3. Add sample data

Employees

ID	Code	FirstName	LastName	Birthday	Sex	Address	Salary	SupervisorID	DepartmentID
1	123456789	John	Smith	1965-01-09	M	731 Fondren, Houston, TX	30000	2	5
2	333445555	Franklin	Wong	1955-12-08	M	638 Voss, Houston, TX	40000	8	5
3	999887777	Alicia	Zelaya	1968-01-19	F	3321 Castle, Spring, TX	25000	4	4
4	987654321	Jennifer	Wallace	1941-06-20	F	291 Berry, Bellaire, TX	43000	8	4
5	666884444	Ramesh	Narayan	1962-09-15	M	975 Fire Oak, Humble, TX	38000	2	5
6	453453453	Joyce	English	1972-07-31	F	5631 Rice, Houston, TX	25000	2	5
7	987987987	Ahmad	Jabbar	1969-03-29	M	980 Dallas, Houston, TX	25000	4	4
8	888665555	James	Borg	1937-11-10	M	450 Stone, Houston, TX	50000	NULL	1

Departments

ID	Name	ManagerID	ManagerStartDate
5	Research	2	1988-05-22
4	Administration	4	1995-01-01
1	Headquarters	8	1981-06-19

DepartmentLocations

DepartmentID	Location
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Projects

ID	Name	Location	DepartmentID
1	ProductX	Bellaire	5
2	ProductY	Sugarland	5
3	ProductZ	Houston	5
10	Computerization	Stafford	4
20	Reorganization	Houston	1
30	Newbenefits	Stafford	4

EmployeeProjects

EmployeeID	ProjectID	Hours
1	1	32.5
1	2	7.5
5	3	40.0
6	1	20.0
6	2	20.0
2	2	10.0
2	3	10.0
2	10	10.0
2	20	10.0
3	30	30.0
3	10	10.0
7	10	35.0
7	30	20.0
4	30	20.0
4	20	15.0
8	20	NULL

Dependents

EmployeeID	Name	Sex	Birthday	Relationship
2	Alice	F	1986-04-05	Daughter
2	Theodore	M	1983-10-25	Son
2	Joy	F	1958-05-03	Spouse
4	Abner	M	1942-02-28	Spouse
1	Michael	M	1988-01-04	Son
1	Alice	F	1988-12-30	Daughter
1	Elizabeth	F	1967-05-05	Spouse

4. Write queries (comment on the query number and requirement):
- **Q1:** Retrieve all distinct salary values of employees

- **Q2:** Retrieve all employees whose address is in Houston, Texas
 - **Q3:** Make a list of all project numbers for projects that involve an employee whose last name is 'Smith', EITHER as a worker or as a manager of the department that controls the project
 - **Q4:** Remove all dependents of employee ID 4
 - **Q5:** Retrieve the names of all employees who do not have supervisors
 - **Q6:** Rewrite Q3 but using nested query
 - **Q7:** Retrieve the names of employees who have no dependents
5. Remove database (clean up)

2. Homework

H: YY: 05 (5 queries)

Database name: Company (Ref: Fundamentals of database systems)

Create SQL file with requirements:

1. Create database
2. Create tables
3. Add sample data
4. Write queries (comment query number and requirement):
 - **Q11:** Retrieve the names of all employees who work in the department that has the employee with the highest salary among all employees
 - **Q12:** Retrieve the names of all employees whose supervisor's supervisor has ID 8
 - **Q13:** Retrieve the Social Security numbers of all employees who work on project ID 1, 2, or 3
 - **Q14:** Retrieve the name of each employee who has a dependent with the same first name and is the same sex as the employee
 - **Q15:** List the names of managers who have at least one dependent
5. Remove database (clean up)